

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

**Product Description:** Phosphate-buffered saline (PBS, 10X), with Triton® X-100  
**Cat No. :** J63521

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

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## Physical hazards

Based on available data, the classification criteria are not met

## Health hazards

Serious Eye Damage/Eye Irritation

Category 2 (H319)

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

## Hazard Statements

H319 - Causes serious eye irritation

## Precautionary Statements

P280 - Wear eye protection/ face protection  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P337 + P313 - If eye irritation persists: Get medical advice/attention

## 2.3. Other hazards

Contains a known or suspected endocrine disruptor  
Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties  
Contains a substance on the National Authorities Endocrine Disruptor Lists

## Section 3: Composition/information on ingredients

### 3.2. Mixtures

| Component                                                                                    | CAS No    | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                    |
|----------------------------------------------------------------------------------------------|-----------|-----------|----------|----------------------------------------------------------------------|
| Water                                                                                        | 7732-18-5 | 231-791-2 | 89.38    | -                                                                    |
| Sodium chloride                                                                              | 7647-14-5 | 231-598-3 | 8        | -                                                                    |
| Phosphoric acid, disodium salt                                                               | 7558-79-4 | 231-448-7 | 1.42     | -                                                                    |
| Poly(oxy-1,2-ethanediyl),<br>alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>omega.-hydroxy- | 9002-93-1 |           | 1        | Acute Tox. 4 (H302)<br>Eye Dam. 1 (H318)<br>Aquatic Chronic 2 (H411) |
| Potassium chloride                                                                           | 7447-40-7 | 231-211-8 | 0.2      | -                                                                    |

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Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## Section 5: Firefighting measures

### 5.1. Extinguishing media

**Suitable Extinguishing Media**  
Not combustible.

**Extinguishing media which must not be used for safety reasons**  
No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**  
Hydrogen chloride, Oxides of phosphorus, Potassium oxides, Sodium oxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6: ACCIDENTAL RELEASE MEASURES

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## 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

## 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

## 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 12

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### **Exposure limits**

List source(s):

| Component          | Bulgaria                   | Croatia | Ireland | Cyprus | Czech Republic |
|--------------------|----------------------------|---------|---------|--------|----------------|
| Potassium chloride | TWA: 5.0 mg/m <sup>3</sup> |         |         |        |                |

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| Component          | Latvia                   | Lithuania                     | Luxembourg | Malta | Romania |
|--------------------|--------------------------|-------------------------------|------------|-------|---------|
| Sodium chloride    | TWA: 5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> IPRD |            |       |         |
| Potassium chloride | TWA: 5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> IPRD |            |       |         |

| Component                      | Russia                    | Slovak Republic | Slovenia | Sweden | Turkey |
|--------------------------------|---------------------------|-----------------|----------|--------|--------|
| Sodium chloride                | MAC: 5 mg/m <sup>3</sup>  |                 |          |        |        |
| Phosphoric acid, disodium salt | MAC: 10 mg/m <sup>3</sup> |                 |          |        |        |
| Potassium chloride             | MAC: 5 mg/m <sup>3</sup>  |                 |          |        |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                               | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-----------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Sodium chloride<br>7647-14-5 ( 8 )      |                              | DNEL = 295.52mg/kg bw/day       |                                | DNEL = 295.52mg/kg bw/day         |
| Potassium chloride<br>7447-40-7 ( 0.2 ) |                              | DNEL = 910mg/kg bw/day          |                                | DNEL = 303mg/kg bw/day            |

| Component                               | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-----------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Sodium chloride<br>7647-14-5 ( 8 )      |                                  | DNEL = 2068.62mg/m <sup>3</sup>     |                                    | DNEL = 2068.62mg/m <sup>3</sup>       |
| Potassium chloride<br>7447-40-7 ( 0.2 ) |                                  | DNEL = 5320mg/m <sup>3</sup>        |                                    | DNEL = 1064mg/m <sup>3</sup>          |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                            | Fresh water     | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|------------------------------------------------------|-----------------|----------------------|--------------------|------------------------------------|--------------------------|
| Sodium chloride<br>7647-14-5 ( 8 )                   | PNEC = 5mg/L    |                      |                    | PNEC = 500mg/L                     | PNEC = 4.86mg/kg soil dw |
| Phosphoric acid, disodium salt<br>7558-79-4 ( 1.42 ) | PNEC = 0.05mg/L |                      | PNEC = 0.5mg/L     | PNEC = 50mg/L                      |                          |
| Potassium chloride<br>7447-40-7 ( 0.2 )              | PNEC = 0.1mg/L  |                      | PNEC = 1mg/L       | PNEC = 10mg/L                      |                          |

| Component                      | Marine water     | Marine water sediment | Marine water Intermittent | Food chain | Air |
|--------------------------------|------------------|-----------------------|---------------------------|------------|-----|
| Phosphoric acid, disodium salt | PNEC = 0.005mg/L |                       |                           |            |     |

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|                                         |                |  |  |  |  |
|-----------------------------------------|----------------|--|--|--|--|
| 7558-79-4 ( 1.42 )                      |                |  |  |  |  |
| Potassium chloride<br>7447-40-7 ( 0.2 ) | PNEC = 0.1mg/L |  |  |  |  |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

#### Physical State

Liquid

#### Appearance

Colorless

#### Odor

No information available

#### Odor Threshold

No data available

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

#### Flammability (liquid)

No data available

#### Flammability (solid,gas)

Not applicable

Liquid

#### Explosion Limits

No data available

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|                                                                                                |                          |                                   |
|------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------|
| Flash Point                                                                                    | No information available | Method - No information available |
| Autoignition Temperature                                                                       | No data available        |                                   |
| Decomposition Temperature                                                                      | No data available        |                                   |
| pH                                                                                             | 7.4                      |                                   |
| Viscosity                                                                                      | No data available        |                                   |
| Water Solubility                                                                               | Miscible                 |                                   |
| Solubility in other solvents                                                                   | No information available |                                   |
| Partition Coefficient (n-octanol/water)                                                        |                          |                                   |
| Component                                                                                      | log Pow                  |                                   |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phe<br>nyl]-.omega.-hydroxy- | 2.7                      |                                   |
| Vapor Pressure                                                                                 | No data available        |                                   |
| Density / Specific Gravity                                                                     | No data available        |                                   |
| Bulk Density                                                                                   | Not applicable           | Liquid                            |
| Vapor Density                                                                                  | No data available        | (Air = 1.0)                       |
| Particle characteristics                                                                       | Not applicable (liquid)  |                                   |

## 9.2. Other information

## Section 10: Stability and reactivity

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization

No information available.

#### Hazardous Reactions

None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

Oxidizing agent.

### 10.6. Hazardous decomposition products

Hydrogen chloride. Oxides of phosphorus. Potassium oxides. Sodium oxides.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

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| Component                                                                                      | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation            |
|------------------------------------------------------------------------------------------------|---------------------------|-------------------------------|----------------------------|
| Water                                                                                          | -                         | -                             | -                          |
| Sodium chloride                                                                                | LD50 = 3550 mg/kg ( Rat ) | LD50 > 10000 mg/kg ( Rabbit ) | LC50 > 42 mg/L ( Rat ) 1 h |
| Phosphoric acid, disodium salt                                                                 | LD50 = 17 g/kg ( Rat )    | -                             | -                          |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | 1800 mg/kg ( Rat )        | -                             | -                          |
| Potassium chloride                                                                             | LD50 = 2600 mg/kg ( Rat ) | -                             | -                          |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

Ecotoxicity effects

| Component       | Freshwater Fish                      | Water Flea          | Freshwater Algae |
|-----------------|--------------------------------------|---------------------|------------------|
| Sodium chloride | Pimephals prome: LC50: 7650 mg/L/96h | EC50: 1000 mg/L/48h |                  |

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|                                                                                                |                                                                                                    |                    |                     |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------|---------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | LC50 = 8.9 mg/L 96H<br>LC50 = 4.0 mg/l 96H<br>(Pimephales promelus)                                | EC50 = 26 mg/L 48h | -                   |
| Potassium chloride                                                                             | Lepomis macrochirus: LC50:<br>1060 mg/L /96h<br>Pimephales promelas: LC50: 750<br>- 1020 mg/L /96h | EC50: 825 mg/L/48h | EC50: 2500 mg/L/72h |

| Component                                                                                      | Microtox | M-Factor |
|------------------------------------------------------------------------------------------------|----------|----------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | -        |          |

## 12.2. Persistence and degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

| Component                                                                                                     | Degradability |
|---------------------------------------------------------------------------------------------------------------|---------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy-<br>9002-93-1 ( 1 ) | 60% >28 days  |

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component                                                                                      | log Pow | Bioconcentration factor (BCF) |
|------------------------------------------------------------------------------------------------|---------|-------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | 2.7     | No data available             |

## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

## 12.5. Results of PBT and vPvB assessment

No data available for assessment.

## 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information**  
**Assess endocrine disrupting properties for the environment**

Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605. Contains a substance on the National Authorities Endocrine Disruptor Lists.

| Component                                                                                  | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|--------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy- | Group III Chemical                       | -                                                |

| Component                                                                                                     | EU National Authorities Endocrine Disruptor Lists - Environment | Japan - Endocrine Disruptor Information |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy-<br>9002-93-1 ( 1 ) | List I                                                          | -                                       |

## 12.7. Other adverse effects

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13: Disposal considerations

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## 13.1. Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                                                                                                                                                              |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## Section 14: Transport information

IMDG/IMO Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

ADR Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

IATA Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

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| Component                                                                                      | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|------------------------------------------------------------------------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Water                                                                                          | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |
| Sodium chloride                                                                                | 7647-14-5 | 231-598-3 | -      | -   | X     | X    | KE-31387 | X    | X    |
| Phosphoric acid, disodium salt                                                                 | 7558-79-4 | 231-448-7 | -      | -   | X     | X    | KE-12344 | X    | X    |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)<br>phenyl]-.omega.-hydroxy- | 9002-93-1 | -         | -      | -   | X     | X    | KE-33568 | X    | X    |
| Potassium chloride                                                                             | 7447-40-7 | 231-211-8 | -      | -   | X     | X    | KE-29086 | X    | X    |

| Component                                                                                      | CAS No    | TSCA | TSCA Inventory<br>notification -<br>Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------------------------------------------------------------------------------------|-----------|------|-----------------------------------------------------|-----|------|------|-------|-------|
| Water                                                                                          | 7732-18-5 | X    | ACTIVE                                              | X   | -    | X    | X     | X     |
| Sodium chloride                                                                                | 7647-14-5 | X    | ACTIVE                                              | X   | -    | X    | X     | X     |
| Phosphoric acid, disodium salt                                                                 | 7558-79-4 | X    | ACTIVE                                              | X   | -    | X    | X     | X     |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)<br>phenyl]-.omega.-hydroxy- | 9002-93-1 | X    | ACTIVE                                              | X   | -    | X    | X     | X     |
| Potassium chloride                                                                             | 7447-40-7 | X    | ACTIVE                                              | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                                                                                      | CAS No    | REACH (1907/2006) -<br>Annex XIV - Substances<br>Subject to Authorization                                                                                                                                                                                                                                                                                                                                                        | REACH (1907/2006) -<br>Annex XVII - Restrictions<br>on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 -<br>Candidate List of<br>Substances of Very High<br>Concern (SVHC)                            |
|------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Water                                                                                          | 7732-18-5 | -                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                                      | -                                                                                                                                            |
| Sodium chloride                                                                                | 7647-14-5 | -                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                                      | -                                                                                                                                            |
| Phosphoric acid, disodium salt                                                                 | 7558-79-4 | -                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                                      | -                                                                                                                                            |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)<br>phenyl]-.omega.-hydroxy- | 9002-93-1 | Endocrine disrupting<br>properties (Article 57(f) -<br>environment)<br>Application date: July 4,<br>2019<br>Sunset date: January 4,<br>2021<br>Exemption - extended<br>latest application and<br>sunset date for the<br>research, development and<br>production of medicinal<br>products or medical<br>devices in view of their use<br>for the diagnosis, treatment<br>or prevention of the<br>coronavirus disease<br>(COVID-19) | -                                                                                      | SVHC Candidate list -<br>Equivalent level of concern<br>having probable serious<br>effects to the environment<br>(Article 57f - environment) |
| Potassium chloride                                                                             | 7447-40-7 | -                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                                      | -                                                                                                                                            |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Major Accident<br>Notification | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Safety Report<br>Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Water     | 7732-18-5 | Not applicable                                                                                  | Not applicable                                                                                 |

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|                                                                                         |           |                |                |
|-----------------------------------------------------------------------------------------|-----------|----------------|----------------|
| Sodium chloride                                                                         | 7647-14-5 | Not applicable | Not applicable |
| Phosphoric acid, disodium salt                                                          | 7558-79-4 | Not applicable | Not applicable |
| Poly(oxy-1,2-ethanediyl), .alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy- | 9002-93-1 | Not applicable | Not applicable |
| Potassium chloride                                                                      | 7447-40-7 | Not applicable | Not applicable |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**

Not applicable

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 1 (self classification)

| Component                                                                               | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------------------------------------------------------------------------------|---------------------------------------|-------------------------|
| Sodium chloride                                                                         | WGK1                                  |                         |
| Phosphoric acid, disodium salt                                                          | WGK1                                  |                         |
| Poly(oxy-1,2-ethanediyl), .alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy- | WGK2                                  |                         |
| Potassium chloride                                                                      | WGK1                                  |                         |

| Component          | France - INRS (Tables of occupational diseases)      |
|--------------------|------------------------------------------------------|
| Sodium chloride    | Tableaux des maladies professionnelles (TMP) - RG 78 |
| Potassium chloride | Tableaux des maladies professionnelles (TMP) - RG 67 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                                                                                                  | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Sodium chloride<br>7647-14-5 ( 8 )                                                                         | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |
| Poly(oxy-1,2-ethanediyl), .alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy-<br>9002-93-1 ( 1 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

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## Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H318 - Causes serious eye damage  
H319 - Causes serious eye irritation  
H411 - Toxic to aquatic life with long lasting effects

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

## Prepared By

Health, Safety and Environmental Department

## Revision Date

30-Nov-2024

## Revision Summary

Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**